# Ryan Keith

**Software Engineer · Full-Stack, Geospatial & AI Systems**

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## SUMMARY


Software engineer at Amgraf, Inc. since February 2024, writing code since age 14. Builds Python backend systems: data pipelines (FastAPI, Celery, PostgreSQL, AWS S3), LLM-powered tools (LangChain, Claude and OpenAI APIs) and internal IT automation (Django, Jira, Slack). Recent portfolio work adds full-stack geospatial apps: React + TypeScript frontends, FastAPI services running spatial SQL on PostGIS, and CI that tests against a real database. B.S. in Information Technology from the University of Kansas with a 4.0 GPA; Eagle Scout.


## EXPERIENCE

**Software Engineer · Amgraf, Inc.** Kansas City, MO · Feb 2024 – Present

- Build Python backend services and data pipelines with FastAPI, Celery, PostgreSQL and AWS S3.
- Develop LLM-powered internal tools with LangChain and the Claude and OpenAI APIs.
- Automate internal IT workflows with Django, the Jira REST API and Slack integrations.

**Student Technician · Johnson County Community College** Overland Park, KS · Oct 2019 – Oct 2022

- Resolved student technical issues by phone and live chat, walking users through each fix step by step.
- Built a Python tool that inserts commonly sent chat replies, cutting typos and speeding up live-chat support.
- Automated a documentation ticket for every support interaction, saving technicians time after each session.
- Presented a larger application to JCCC's full IT department.

## SKILLS

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|---|---|
| **Languages** | Python, TypeScript, JavaScript, SQL, HTML/CSS |
| **Frameworks** | FastAPI, Django, React, Vite, Pydantic |
| **Data & GIS** | PostgreSQL, PostGIS (spatial SQL, GiST indexes, KNN, hexagon grids), Redis, MapLibre GL JS, Turf.js, OpenStreetMap data |
| **AI & ML** | Claude API (tool calling), OpenAI API, LangChain, Whisper speech-to-text, sentiment analysis, satellite image change detection |
| **Infrastructure** | Docker, Celery, AWS S3, GitHub Actions CI, Render, Netlify, GitHub Pages |
| **Testing** | pytest (incl. PostGIS integration tests), Vitest, Playwright |
| **Integrations** | Stripe, SendGrid, Jira REST API, Slack webhooks, ReportLab |
| **Professional** | Technical support, user communication, leadership, planning |

## PORTFOLIO PROJECTS

Live demos and source for every project: musical-marshmallow-0f8bd3.netlify.app · github.com/ryan12keith-eng/portfolio · 89 automated tests run in GitHub Actions

**GeoMind AI · Smart City GIS Platform** Flagship — React, TypeScript, MapLibre, FastAPI, PostGIS, Claude API

- Built a 3D map of central Kansas City with 441 real OpenStreetMap facilities and rivers, and a GIS assistant that answers plain-language questions.
- FastAPI service runs parameterized PostGIS SQL: radius search, KNN nearest facility, school-coverage gaps on a hexagon grid, density and river proximity.
- Assistant routes questions with Claude tool calling (strict schemas) when a key is configured, otherwise a rule-based router; answers always come from the database and show the SQL that ran.
- Validated inputs and rate limiting; pytest suite runs against a PostGIS container in CI; frontend falls back to in-browser Turf.js when the API sleeps.

**LandScope AI · Land Suitability Scoring** — React, FastAPI, PostGIS, PostgreSQL

- Measured 415 real land-use sites in a PostGIS materialized view: true area and geodesic distance to the nearest major road, school and hospital.
- Ranked sites with configurable weights (scoring rules mirrored and tested in Python and TypeScript); saved site reports in PostgreSQL with shareable links.

**EarthWatch AI · Flood Change Detection** TypeScript, Canvas image analysis

- Compared NASA Landsat images of the 2024 Rufiji River flood pixel by pixel to detect newly flooded land (~11% of the scene), classify land cover and compute a vegetation index.

**Workflow apps · PrintFlow, Smart Quote Assistant, Job Application Tracker** React, TypeScript

- Print job and proof portal with role-based permissions, an enforced status workflow, validated uploads and an audit trail; quote-email spec extraction with deterministic, unit-tested pricing; drag-and-drop application tracker with posting analysis.

## EARLIER PROJECTS

**Print-Job Billing & Invoicing Engine** Python, FastAPI, Celery, Redis, PostgreSQL, Stripe, AWS S3

- Built an ingestion pipeline where Celery workers watch an S3 bucket and process uploaded PDF and PostScript print jobs in the background.
- Wrote a parser that extracts page count, dimensions, color profiles and vector complexity to estimate each job's cost.
- Designed a pricing API driven by a per-client pricing matrix that changes without a redeploy.
- Integrated Stripe and SendGrid to generate invoices, email clients and export accounting manifests automatically.

**AI Customer Support QA Engine** Python, LangChain, Claude/OpenAI API, Whisper, React, PostgreSQL

- Built a batch pipeline that ingests chat logs (JSON/CSV) and transcribes support calls with Whisper.
- Used an LLM to audit each conversation against support guidelines (greeting, identity verification, accuracy) and score technician compliance.
- Added sentiment analysis that flags rising customer frustration or wrong answers; built a React dashboard of resolution time, sentiment trends and technician scores.

**IT Asset Provisioning & Inventory System** Python, Django, PostgreSQL, Jira REST API, Slack, ReportLab, Docker

- Designed a schema tracking hardware serial and MAC addresses, purchase and warranty dates, and assigned employees.
- Automated onboarding so each new-hire event opens a Jira ticket with a hardware setup checklist; scheduled stock checks alert purchasing in Slack.
- Generated weekly ReportLab PDF reports on asset depreciation and assignments; containerized the app with Docker.

## EDUCATION & LEADERSHIP

**B.S., Information Technology · The University of Kansas** Aug 2021 – May 2023 · GPA 4.0

**A.S., Information Technology · Johnson County Community College** 2019 – 2021

- **Eagle Scout**, Boy Scouts of America: earned at 17 after five years of leadership and service projects.
- **Full-time volunteer missionary**, British Columbia, Canada: two years of daily outreach, which built communication and self-directed planning skills.